

Climate Health Vulnerability Cluster Analysis Report

Rochester, NY Census Tracts

By

Climate and Health Analytics Division

Executive Summary

Four distinct vulnerability clusters among the census tracts. **Cluster 4** ($n = 21$ tracts) represents the highest-risk group, driven by a combination of structural and socioeconomic disadvantage, including poverty, transit dependence, limited English proficiency, no-vehicle housing, and high child vulnerability. **Cluster 1** ($n = 33$ tracts) reflects a different high-risk profile characterized by aging populations, elevated disability, and moderate socioeconomic stress. **Cluster 3** ($n = 19$) is a distinct moderate-high risk group primarily defined by economic high poverty. **Cluster 2** ($n = 58$) is the lowest-risk and most socioeconomically stable cluster.

1 Clustered Vulnerability Patterns

The cluster heatmaps show clear separation in vulnerability indicators across census tracts. High values are concentrated in specific groups rather than distributed uniformly across the city, indicating spatial clustering of disadvantage. Cluster 4 stands out as the most vulnerable group, with elevated structural barriers including limited English proficiency, housing strain, and child poverty. Cluster 3 shows strong economic deprivation, particularly in poverty and unemployment. Cluster 1 is distinguished by higher aging population, disability, and health burden rather than extreme economic distress. Cluster 2 appears comparatively advantaged cluster.

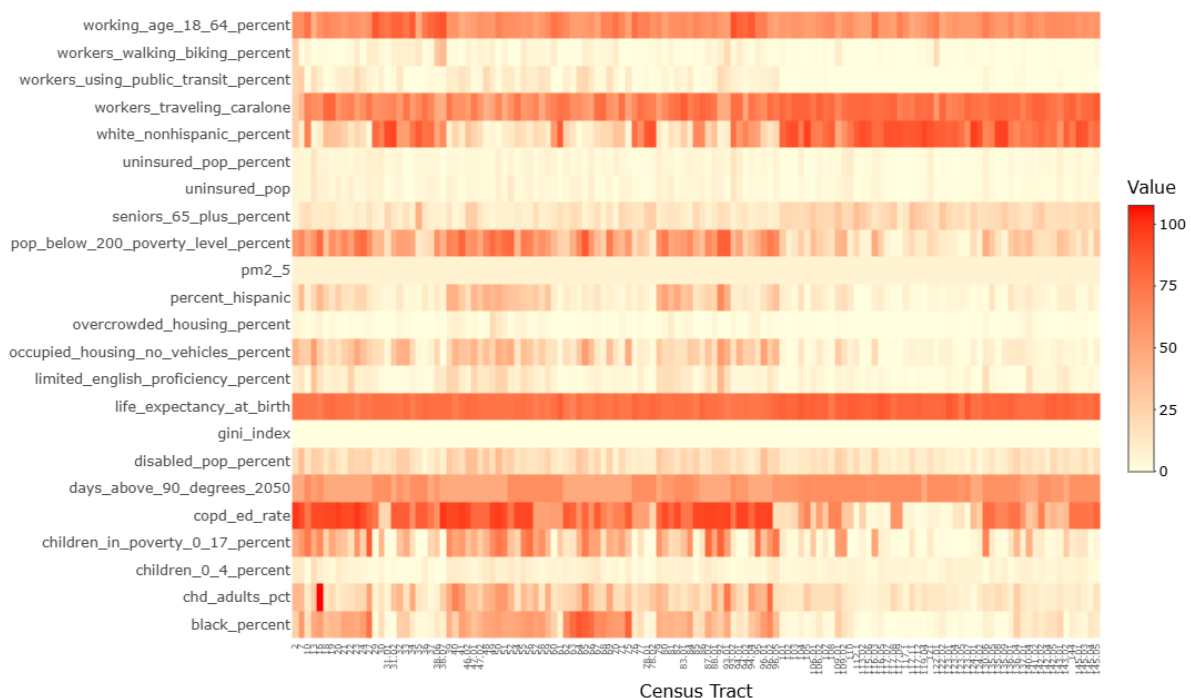


Figure 1: Heatmap of all census tracts showing the distribution of vulnerability indicators across clusters.

1.1 PCA and Cluster Separation

The PCA plot shows clear separation among the clusters, indicating that the four groups represent distinct combinations of neighborhood characteristics. Cluster 4 is farthest along the high-vulnerability direction, while Cluster 2 lies on the opposite end, representing low vulnerability. Cluster 3 and Cluster 1 occupy intermediate but distinct positions. This pattern suggests that PC1 primarily reflects a socioeconomic gradient (advantage versus disadvantage), while PC2 captures variation related to demographic structure, aging, and housing or health-related stress. The separation between Cluster 3 and Cluster 4 further indicates that economic hardship and structural vulnerability represent different dimensions of risk.

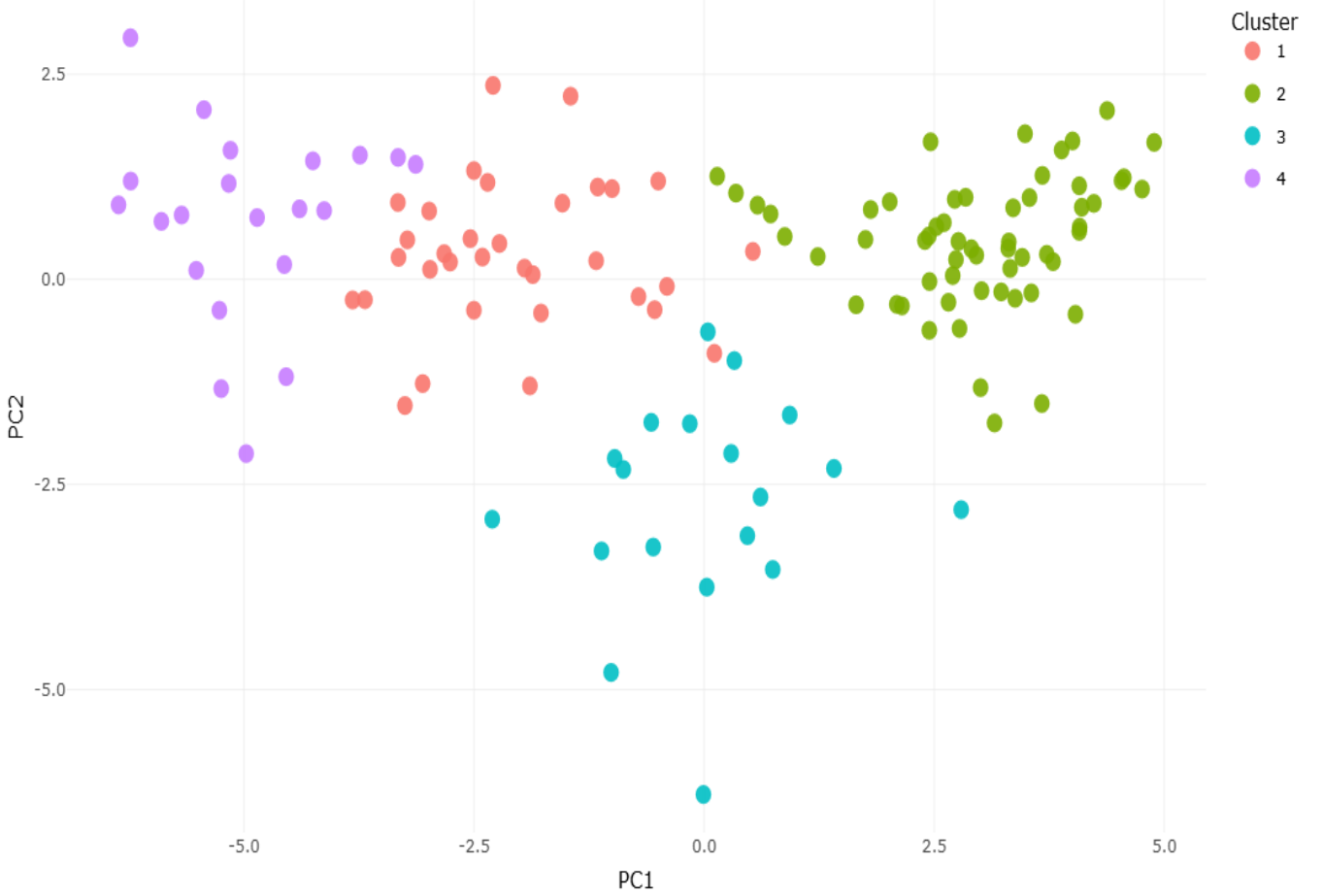
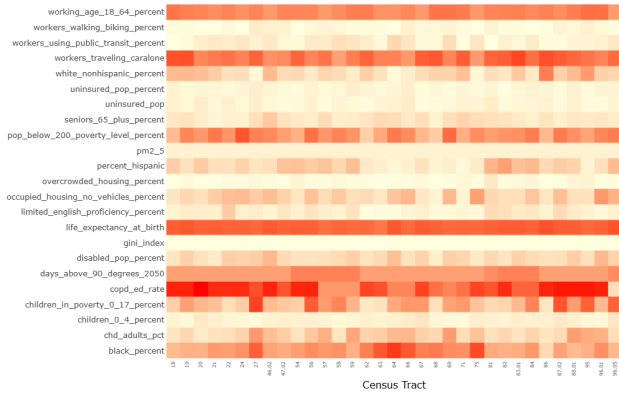


Figure 2: PCA plot with k-means clusters showing separation among census tract profiles.

1.2 Cluster Comparison

The four cluster heatmaps show distinct vulnerability patterns. Cluster 4 has the strongest concentration of multidimensional risk, including structural barriers such as limited English proficiency, overcrowding, and child poverty. Cluster 3 is primarily defined by economic vulnerability, with high poverty, unemployment, and public assistance dependence. Cluster 1 shows a different risk profile, with higher concentrations of elderly population, disability, and health burden alongside moderate socioeconomic stress. Cluster 2 shows the lowest composite risk and the most stable overall pattern.



1.5 Cluster Characteristics and Vulnerability Profiles

Table 1: Vulnerability Cluster Profiles

Cluster	Size	Mean Risk	Key Characteristics
4 (Highest)	21 tracts	0.543	Highest composite risk, strongest poverty and uninsured burden, limited English proficiency, transit dependence, and no-vehicle housing.
1 (High)	33 tracts	0.145	Elevated poverty and child poverty, higher Black population share, and moderate overall health burden.
3(Moderate High)	19 tracts	0.028	Mixed socioeconomic and mobility stressors, with working-age concentration, walking/biking, and overcrowding patterns.
2 (Low)	58 tracts	-0.288	Lowest composite risk, lower poverty, lower uninsured rates, higher White non-Hispanic share, and higher life expectancy.

1.5.1 Cluster 4: Highest Risk Profile

Cluster 4 Profile

- **21 tracts** – highest priority for intervention.
- Strongest concentration of multidimensional deprivation.
- **Highest poverty burden and greatest transit dependence.**
- **High Hispanic concentration and elevated limited English proficiency.**
- **High no-vehicle housing, child poverty, and uninsured burden.**
- Represents deeply entrenched structural disadvantage.

1.5.2 Cluster 1: High-Risk Profile

Cluster 1 Profile

- **33 tracts** with elevated composite risk.
- Stronger poverty burden than the lower-risk groups.
- Elevated Black population share and child poverty.
- Moderate health burden and socioeconomic strain.
- Represents a high-risk group requiring targeted support.

1.5.3 Cluster 3: Moderate-High Risk Profile

Cluster 3 Profile

- **19 tracts** with moderate-high composite risk.
- Distinct pattern shaped by working-age concentration.
- Elevated walking/biking behavior and uninsured residents.
- Some overcrowding and health stress.
- Represents a separate vulnerability pathway.

1.5.4 Cluster 2: Low Risk Profile

Cluster 2 Profile

- **58 tracts** with the lowest composite risk.
- Most stable socioeconomic profile across all clusters.
- Lower poverty burden and comparatively strong life expectancy.
- Lower uninsured rates, lower overcrowding, and fewer access barriers.
- Serves as the baseline reference group for vulnerability comparison.

1.6 Key Actionable Insights

Table 2: Priority Matrix

Priority	Cluster	Action Required
1: Immediate	Cluster 4 (21 tracts)	Severe multidimensional deprivation with high poverty, transit dependence, no-vehicle housing, and language access barriers.
2: Urgent	Cluster 1 (33 tracts)	Elevated poverty and socioeconomic strain requiring targeted stabilization.
3: Monitor	Cluster 3 (19 tracts)	Moderate-high risk group with distinct demographic, housing, and health stress patterns.
4: Baseline	Cluster 2 (58 tracts)	Lowest risk group representing more stable socioeconomic conditions.

Final Interpretation

Cluster 4 represents the highest concentration of vulnerability and should be prioritized for immediate intervention. Cluster 1 follows as a high-risk group requiring targeted support. Cluster 3 shows moderate-high risk with distinct housing and demographic stress patterns, while Cluster 2 functions as the baseline low-risk reference group.

Technical Notes

Report generated from PCA and k-means analysis of 23 socioeconomic and health indicators across Rochester census tracts. Cluster-level means, tract rankings, and driver scores were used to update the interpretation. The PCA plot confirms strong separation among the four clusters, especially between the highest-risk and lowest-risk groups.

2 Appendix

2.1 Census Tracts by Cluster

Table 3: Census Tracts by Cluster

Cluster	Census Tracts
1	96, 84, 81, 70, 66, 65, 64, 63, 60, 58, 56, 55, 51, 50, 49, 48, 47.02, 45, 44, 42, 41, 39, 38.06, 37, 35, 32, 30, 28, 27, 23, 21, 19, 18, 17, 14
2	123.05, 119.04, 115.06, 117.09, 124.02, 117.10, 123.04, 113.02, 102, 31.01, 117.07, 117.12, 117.08, 123.07, 124.01, 103, 76, 141.02, 122.02, 117.11, 142.04, 135.08, 145.04, 101, 123.01, 112.10, 78.02, 108, 78.01, 61, 137.01, 144, 142.05, 135.09, 105, 142.02, 116.05, 107, 145.01, 143.02, 36, 121, 62, 145.03, 106.02, 106.01, 110, 71, 54, 145.05, 34, 40, 109.02, 109.01, 136.04, 143.01, 104, 136.01, 116.03
3	37, 122.01, 94.02, 77, 74, 72, 85, 29, 10, 94.04, 31.02, 38.07, 70, 32, 93.02, 30, 33, 43, 46, 25, 24, 22, 53, 57, 59, 60, 89, 94.01, 106
4	4, 5, 7, 8, 9, 10, 11, 12, 13, 15, 16, 20, 26, 29, 31, 33, 41, 48, 50, 52, 80, 96.02, 93.01, 96.01, 47.01, 79, 55, 39, 51, 40, 23, 18